Rational Expressions and Equations

Concept

Rational expressions are polynomial fractions. Original denominators determine restrictions.

Cancel factors

Never cancel separate terms of a sum.

Add / subtract

Use a common denominator, then combine whole numerators.

Divide

Multiply by the reciprocal; the divisor must also be non-zero.

Solve

Record exclusions → multiply by LCD → solve → reject invalid roots.

Example 1

Simplify $(x^2 - 4)/(x^2 + 3x + 2)$.

Solution

1. Factor: $(x - 2)(x + 2)/[(x + 1)(x + 2)]$.
2. Restrictions: $x \neq -1, -2$. Cancel factor $x + 2$.
3. Result: $(x - 2)/(x + 1)$, keeping both restrictions.

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Example 2

Solve $(x + 1)/(x - 1) = 2/(x - 1)$.

Solution

1. Restriction: $x \neq 1$.

2. Multiply by $x - 1$: $x + 1 = 2 \rightarrow x = 1$.

3. Reject excluded value 1. There is no solution.

Rational Expressions and Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify $(x^2 + 5x + 6)/(x + 2)$.

2. Subtract $3/x - 1/(2x)$.

3. Solve $3/(x - 2) = 1$.

4. Solve $x/(x + 1) = 1$. State restrictions in each case.

Rational Expressions and Equations

Practice answers | check after trying

1. $x + 3$, $x \neq -2$.

2. $6/(2x) - 1/(2x) = 5/(2x)$, $x \neq 0$.

3. $x \neq 2$; $3 = x - 2$ gives $x = 5$, which checks.

4. $x \neq -1$; $x = x + 1$ is impossible, so there is no solution.